SPAGHETTILAB | BACKBONE CARRIER

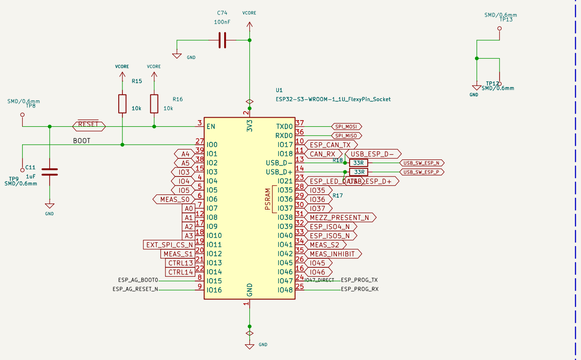
Functional groups | local wires retained | power shown with symbols | cross-block links use named nets

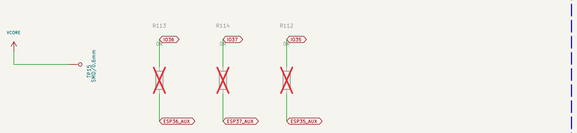
P1–P6 = external experiment rows | A0–A5 = ESP-side signals | J3 = separate direct access. Named arrows connect blocks.

READING GUIDE: Pn = row / connector | Q = power switch | U12/U13 = signal isolation | R56–R61 = current shunts

U1 | OPTIONAL ESP32-S3

Adds Wi-Fi, Bluetooth and familiar ESP32 development tools.

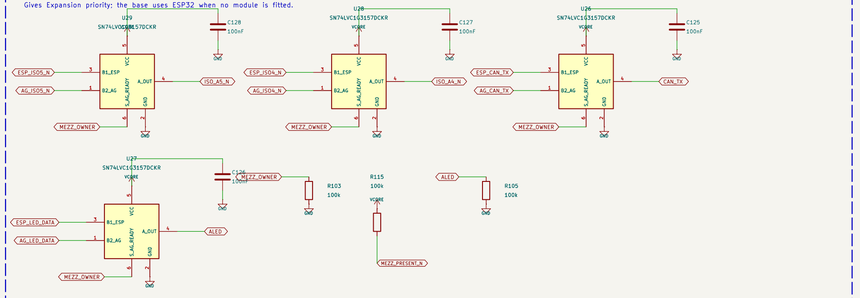




GPIO35/36/37 AUX links are DNP PSRAM guards. Fit only on compatible ESP32 variants. USB does not pass through FPGA logic.

U26–U29 | AUTOMATIC CONTROLLER SELECTION

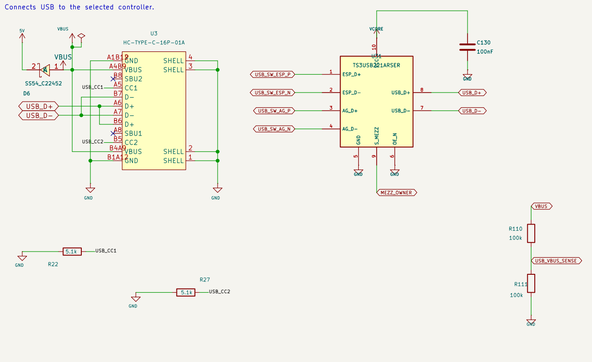
Gives Expansion priority: the base uses ESP32 when no module is fitted.



MEZZ_OWNER LOW = ESP32, HIGH = Expansion. R103 = 100k pull-down: Expansion adds R109 = 10k pull-up. Claim is not firmware-ready. USB, CAN TX, RGB and ISO4/5 are selected.

USB-C + U31 | COMPUTER CONNECTION

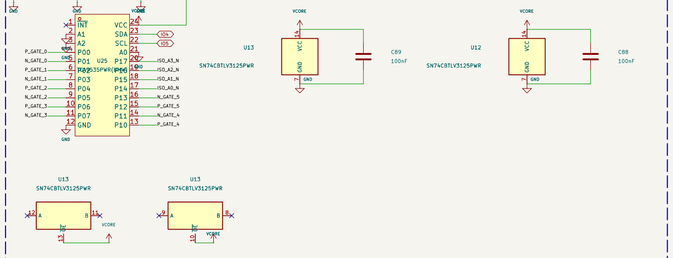
Connects USB to the selected controller.



U31: LOW selects ESP32, HIGH selects Expansion. Both USB series-resistor pairs are fitted. Owner changes require USB detach and re-enumeration.

U25 | SHARED PORT CONFIGURATION

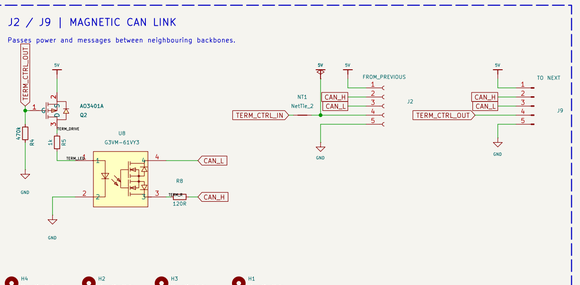
Lets either controller configure the experiment ports over I2C.



U25 and C119 are COMMON: Fit with or without Expansion. One active I2C master at a time. U12/U13 power units and unused channels are shown here.

J2 / J9 | MAGNETIC CAN LINK

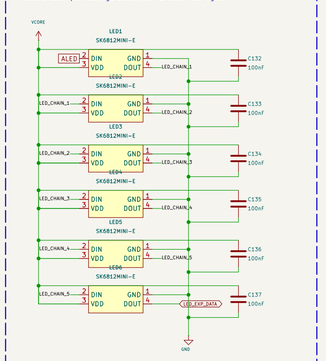
Passes power and messages between neighbouring backbones.



J9 / Q2 / U8 set CAN termination automatically from the physical link. This is independent of MEZZ_OWNER.

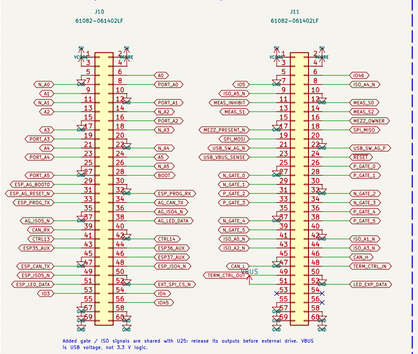
LED1–LED6 | PORT STATUS

Shows each port configuration with its own RGB light.



J10 / J11 | EXPANSION INTERFACE — 2 x 60 CONTACTS

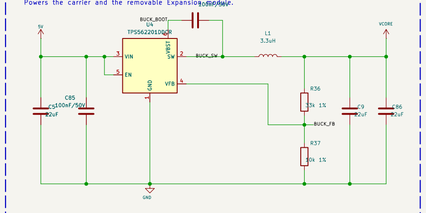
120 contacts: existing services + 22 future-module signals; J11 pins 53 / 54 / 56 remain reserved.



Added gate / ISO signals are shared with U25: release its outputs before external drive. VBUS is USB voltage, not 3.3 V logic.

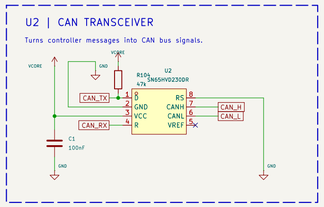
U4 | 5 V TO 3.3 V SUPPLY

Powers the carrier and the removable Expansion module.



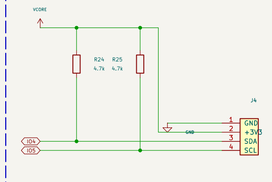
U2 | CAN TRANSCEIVER

Turns controller messages into CAN bus signals.



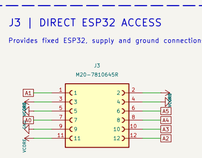
J4 | QWIIC SENSOR LINK

Connects compatible sensors through a four-wire cable.



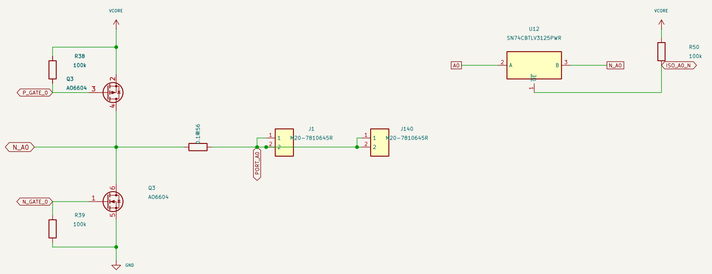
J3 | DIRECT ESP32 ACCESS

Provides fixed ESP32, supply and ground connections.



P1 | CONFIGURABLE EXPERIMENT ROW

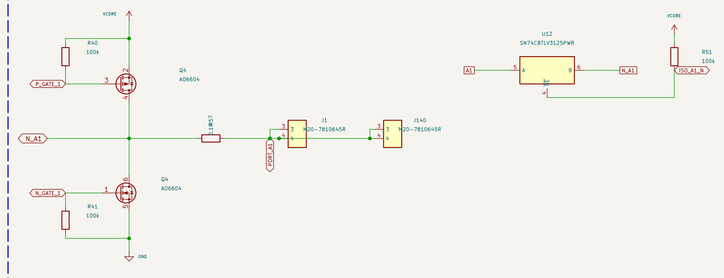
Connects row P1 to its ESP signal, 3.3 V, ground or high impedance.



Same circuit repeats for P1–P6. Never enable P-MOS and N-MOS together. Set GPIO Hi-Z before changing power mode. Shunts stay 0.1 ohm.

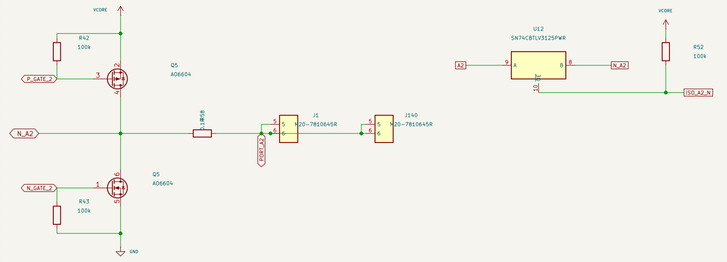
P2 | CONFIGURABLE EXPERIMENT ROW

Connects row P2 to its ESP signal, 3.3 V, ground or high impedance.



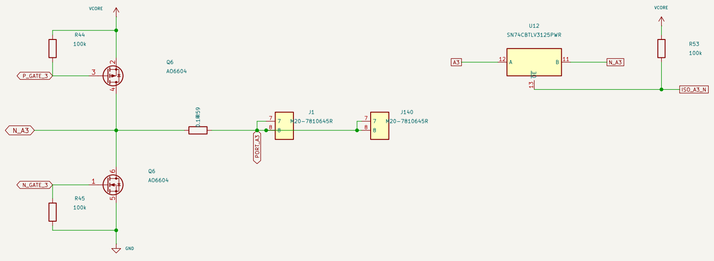
P3 | CONFIGURABLE EXPERIMENT ROW

Connects row P3 to its ESP signal, 3.3 V, ground or high impedance.



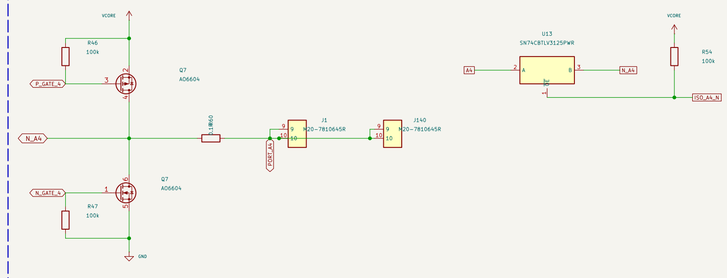
P4 | CONFIGURABLE EXPERIMENT ROW

Connects row P4 to its ESP signal, 3.3 V, ground or high impedance.



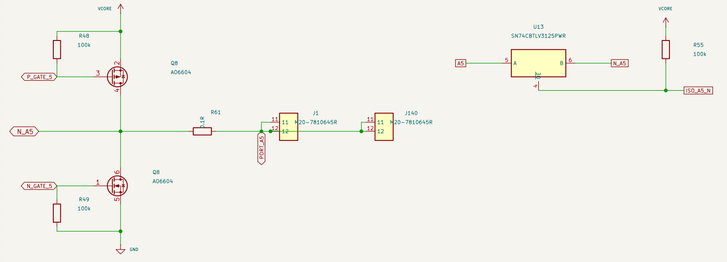
P5 | CONFIGURABLE EXPERIMENT ROW

Connects row P5 to its ESP signal, 3.3 V, ground or high impedance.



P6 | CONFIGURABLE EXPERIMENT ROW

Connects row P6 to its ESP signal, 3.3 V, ground or high impedance.



Functional clusters; wiring preserved. Not production-qualified.

Sheet: /	
File: backbone.kicad_sch	
Title: SpaghettiLAB – Backbone carrier	
Size: User	Date:
KiCad E.D.A. 10.0.3	**Rev: Mezzanine V1 – organ**
	Id: 1/1